

Numerical Linear Algebra First-Year Exam, January 2025

Do 4 (four) problems.

1. Assume $A \in \mathbb{C}^{m \times m}$.

- (a) Show that A has a Schur decomposition.
- (b) If A has a collection of m linearly independent eigenvectors, show that A is diagonalizable.

2. If $q_1, \dots, q_n$ is an orthonormal basis for the subspace $V \subset \mathbb{C}^m$ with $m > n$, prove that the orthogonal projector onto V is QQ^* , where Q is the matrix whose columns are the q_j .

3. Let $A \in \mathbb{C}^{m \times n}$, with $m \geq n$ and $\text{rank}(A) = n \geq 3$. Let $a_1, a_2, \dots$ denote the columns of A .

- (a) Using the modified Gram–Schmidt process, write out expressions for q_1, q_2, q_3 , the first three columns of Q in the QR decomposition of A .
- (b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2 .

4. Let $\|\cdot\|$ be a subordinate (induced) matrix norm.

- (a) If E is a $n \times n$ with $\|E\| < 1$, then show $I + E$ is nonsingular and

$$\|(I + E)^{-1}\| \leq \frac{1}{1 - \|E\|}.$$

- (b) If A is a $n \times n$ invertible and E is $n \times n$ with $\|A^{-1}\|\|E\| < 1$, then show $A + E$ is nonsingular and

$$\|(A + E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|E\|}.$$

5. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m \times n}$. Let u_j denote column j of U .

- (a) Suppose $\text{rank}(A) = p < n < m$. Show $\{u_1, u_2, \dots, u_p\}$ is a basis for $\text{Col}(A)$ and $\{u_{p+1}, u_{p+2}, \dots, u_m\}$ is a basis for $\text{Null}(A^*)$.
- (b) Suppose A is full rank and $x \neq 0$. Let σ_i , $i = 1, \dots, n$, be the singular values of A . Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that it holds.